

GUTERSLOH, Germany

WMO: 103200

Lat: 51.923N

Lon: 8.306E

Elev: 72

StdP: 100.46

Time Zone: 1.00 (EUC)

Period: 97-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-6.2	-12.1	1.3	-7.1	-9.8	1.6	-4.1	12.3	8.6	11.2	8.1	1.2	50	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.8	30.2	19.3	28.0	18.7	26.0	18.0	20.7	27.4	19.8	25.8	18.9	24.2	3.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.6	22.5	17.8	12.9	21.7	17.0	12.2	21.0	59.9	27.6	56.7	26.0	53.9	24.3	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.8	8.4	7.4	-12.6	34.2	3.9	2.1	-15.4	35.7	-17.7	36.9	-19.9	38.1	-22.8	39.6
			-12.9	22.9	3.9	0.8	-15.7	23.5	-18.0	24.0	-20.2	24.4	-23.1	25.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	2.4	2.9	5.6	9.9	13.8	16.6	18.5	18.6	14.9	10.4	6.2	2.6
	DBStd	7.05	4.71	4.40	3.91	3.95	3.51	3.37	3.32	3.19	3.18	3.65	3.49	4.46
	HDD10.0	1031	238	199	142	51	7	0	0	0	1	39	121	232
	HDD18.3	3077	495	431	394	254	145	73	39	35	112	244	365	489
	CDD10.0	1117	1	2	7	47	126	197	263	266	148	53	6	1
	CDD18.3	122	0	0	0	1	6	21	44	43	9	0	0	0
	CDH23.3	1167	0	0	0	13	70	184	426	394	78	2	0	0
	CDH26.7	340	0	0	0	0	12	46	138	132	12	0	0	0
Wind	WSAvg	3.0	3.8	3.5	3.3	2.8	2.7	2.7	2.7	2.4	2.5	3.0	3.1	3.5
Precipitation	PrecAvg	767	58	49	49	62	63	82	80	72	60	55	66	72
	PrecMax	1021	118	119	86	108	121	195	178	174	146	128	105	148
	PrecMin	524	13	15	16	11	24	15	22	31	26	11	28	15
	PrecStd	167	28	28	22	27	24	47	40	32	29	36	21	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	14.6	18.8	25.3	28.2	31.0	33.1	33.9	28.3	22.0	16.2	13.2
		MCWB	10.3	9.9	11.1	15.0	17.5	20.2	20.2	20.3	18.5	15.9	12.4	10.9
	2%	DB	11.5	12.4	16.2	22.1	25.2	27.8	30.4	30.0	25.8	19.4	14.1	11.2
		MCWB	9.6	8.8	10.2	13.8	16.4	19.0	19.5	18.7	18.2	14.9	11.5	9.3
	5%	DB	10.1	10.8	13.9	19.7	23.1	25.2	27.8	27.6	23.0	17.5	12.6	9.8
		MCWB	8.4	8.4	9.7	12.4	15.0	17.8	18.7	18.7	17.0	14.0	10.5	8.3
	10%	DB	8.7	9.2	11.9	17.2	21.1	23.2	25.5	25.1	20.7	15.9	11.2	8.8
		MCWB	7.2	7.2	8.6	11.4	14.5	16.7	18.0	17.9	15.9	13.3	9.3	7.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	10.7	12.7	16.0	19.8	21.3	21.9	22.1	20.2	17.0	13.2	11.4
		MCDB	12.4	13.1	16.9	22.7	24.9	27.5	29.4	29.3	25.5	20.9	14.9	12.8
	2%	WB	10.0	9.7	11.4	14.6	17.9	20.0	20.6	20.6	18.8	15.6	11.9	9.6
		MCDB	11.4	11.7	14.8	20.1	23.0	26.1	27.5	27.4	23.8	18.4	13.8	10.9
	5%	WB	8.5	8.5	10.2	13.3	16.4	18.8	19.6	19.5	17.6	14.6	10.7	8.5
		MCDB	9.9	10.3	13.0	18.1	21.4	23.7	25.8	25.3	22.1	17.1	12.4	9.7
	10%	WB	7.1	7.3	8.9	12.0	14.9	17.6	18.7	18.5	16.4	13.5	9.5	7.2
		MCDB	8.6	9.1	11.6	16.4	19.3	21.8	24.1	23.4	19.9	15.7	10.9	8.5
Mean Daily Temperature Range	5% DB	MDBR	5.4	6.0	7.9	9.8	9.8	9.8	9.6	9.8	9.2	7.8	5.9	4.8
		MCDBR	6.0	8.2	11.9	14.2	13.4	13.1	14.0	13.9	13.0	9.7	7.5	5.3
		MCWBR	4.9	5.5	7.1	7.1	6.4	6.1	5.4	5.5	6.5	5.9	5.3	4.3
	5% WB	MCDBR	5.6	6.3	10.4	12.0	11.6	11.6	12.5	11.7	11.6	8.9	7.0	5.2
		MCWBR	5.0	4.7	6.5	6.5	6.6	5.9	5.2	5.7	6.3	5.9	5.3	4.4
Clear-Sky Solar Irradiance	taub	0.318	0.341	0.373	0.409	0.412	0.412	0.424	0.418	0.388	0.357	0.337	0.315	
	taud	2.384	2.358	2.285	2.226	2.249	2.270	2.245	2.284	2.358	2.427	2.400	2.378	
	Ebn at Noon	685	771	812	825	841	843	824	808	790	744	652	620	
	Edh at Noon	61	82	106	126	129	128	129	118	98	76	60	52	
All-Sky Solar Radiation	RadAvg	0.67	1.35	2.44	3.96	4.80	5.17	4.93	4.18	3.03	1.76	0.80	0.52	
	RadStd	0.09	0.22	0.32	0.47	0.50	0.44	0.65	0.36	0.39	0.17	0.10	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page